

# Helical-Grooved Pulley Dimensioning Guide for Cable-Driven Robot Arm

This document summarizes the fundamental equations and dimensioning approach for designing a helical-grooved cable pulley or drum used in precision cable-driven robotic arms. It includes formulas for torque-tension relationship, friction checks, and helical geometry.

## 1. Torque–Tension Relationship

Equation:  $T_{\text{motor}} = (T_1 - T_2) \times r$

Example: For  $T_{\text{motor}} = 1.0 \text{ N}\cdot\text{m}$  and  $r = 15 \text{ mm} \rightarrow (T_1 - T_2) = 66.7 \text{ N}$ .

## 2. Cable Strength Check

Safety Factor  $\approx 3\text{--}4\times$ .  $F_{\text{rated}} \geq 4 \times T_{\text{max}}$ .

Example: If  $T_{\text{max}} = 63 \text{ N} \rightarrow F_{\text{rated}} \geq 250 \text{ N}$  (1 mm Dyneema  $\approx 300 \text{ N}$ ).

## 3. Capstan / Friction Check

$T_1/T_2 \leq e^{(\mu\theta)}$

Example:  $\mu = 0.15$ ,  $\theta = 4\pi \rightarrow e^{(1.885)} \approx 6.6$ . So  $T_1/T_2 \leq 6.6$  ensures no slip.

## 4. Helical Geometry Calculations

| Parameter    | Symbol    | Formula                        | Example                                    |
|--------------|-----------|--------------------------------|--------------------------------------------|
| Pitch        | $p$       | $1.2 \times d_c$               | 1.2 mm for 1 mm cable                      |
| Groove depth | $h$       | $0.8 \times d_c$               | 0.8 mm                                     |
| Helix angle  | $\lambda$ | $\tan(\lambda) = p / (2\pi r)$ | $\approx 0.73^\circ$ for $r=15 \text{ mm}$ |
| Turns        | $n$       | $(w - 2t_f) / p$               | $\approx 20$ for $w=25 \text{ mm}$         |
| Lead         | $L$       | $p \times n$                   | $\approx 25 \text{ mm total}$              |

## 5. Torque–Angle Transfer

$\theta_{\text{joint}} = (r_{\text{motor}} / r_{\text{joint}}) \times \theta_{\text{motor}}$

Example:  $r_{\text{motor}} = 15 \text{ mm}$ ,  $r_{\text{joint}} = 30 \text{ mm} \rightarrow \text{ratio} = 0.5 \rightarrow \text{torque } \times 2, \text{ speed } \div 2$ .

## Summary of Key Design Steps

Compute torque and tension difference:  $(T_1 - T_2) = T_{\text{motor}} / r$ . Select cable rated  $\geq 4 \times \text{max tension}$ . Check capstan friction condition ( $T_1/T_2 \leq e^{(\mu\theta)}$ ). Design groove with pitch =  $1.2 \times \text{cable } \varnothing$ , depth =  $0.8 \times \text{cable } \varnothing$ . Compute helix angle and turns for desired drum width. Match motor/joint drum radii for desired torque-speed ratio.